

Microsoft offers free online course on Internet of Things

Microsoft has launched a new online course for developers who are looking to venture into the Internet of Things (IoT). The company has announced Azure IoT Developer Specialty training, which is free of cost. This course will allow developers to work with data engineers and other stakeholders to provide business solutions right from the setup of the IoT device to the end of its life cycle.

After completing the course, candidates will know how to

implement Azure services that form an IoT solution, including data analysis, data processing, data storage options, and platform-as-a-service options.

However, to get certified, you need a certain amount of prior knowledge. The official blog states, "Candidates for the Azure IoT Developer Specialty certification should have subject matter expertise developing cloud and edge components of an Azure IoT solution."

Google AI to predict eye condition behind vision loss

Google has developed an artificial intelligence (AI) based system that could help in predicting the progression of an eye condition called wet age-related macular degeneration (AMD). AMD is an eye disease that causes vision loss. It is the third-largest cause of blindness in the world after cataract and glaucoma. Around 75 per cent of patients with AMD have an early form of the disease called 'Dry AMD' that has a relatively mild impact on vision.

A small fraction of patients, however, develop the more sight-threatening form of the disease called 'Wet AMD', wherein abnormal blood vessels develop underneath the retina. These

vessels can leak fluids, which can cause permanent vision loss. This condition affects around fifteen per cent of patients.

Wet AMD affects one eye first, which makes patients heavily reliant on the other eye for their day to day functioning. Now, Google in collaboration with its DeepMind subsidiary and Moorfields Eye Hospital NHS Foundation Trust, has developed an AI model that has the potential to predict whether a patient will develop wet AMD within six months. This system could potentially help doctors plan studies of earlier intervention as well as contribute more broadly to the clinical understanding of the disease and disease progression.

Implantable biosensor that operates without batteries

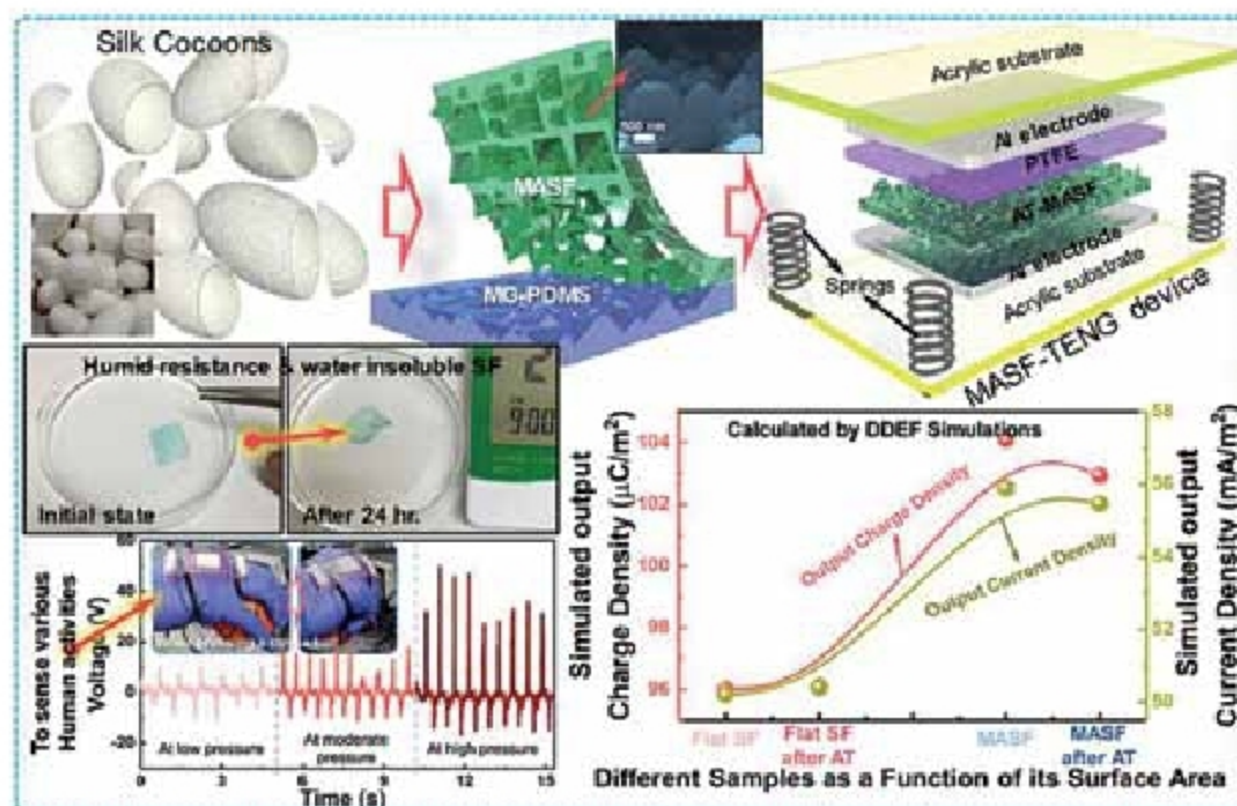
Researchers from the University of Surrey have revealed their new biodegradable motion sensor—paving the way for implanted nanotechnology that could help future sports professionals monitor their movements in a better way to aid rapid improvements.

In a paper published by Nano Energy, a team from Surrey's Advanced Technology Institute (ATI), in collaboration with Kyung Hee University in South Korea, has developed a nano-bio-medical motion sensor, which can be paired with AI systems to recognise movements of distinct body parts. The ATI's technology

is built on its previous work around triboelectric nanogenerators (TENG), where researchers used the technology to harness human movements and generate small amounts of electrical energy.

In their new research, the team from ATI developed a flexible, biodegradable and long-lasting TENG from silk cocoon waste and used a new alcohol treatment technique, which leads to greater durability for the device, even under harsh or humid environments.

Dr Bhaskar Dudem, project lead and research fellow at the ATI says, "We are excited to show the world the immense potential of our durable, silk film based nanogenerator. Its ability to work in severe environments while being able to generate electricity and monitor human movements positions our TENG in a class of its own when it comes to the technology."



The team from the ATI has developed a flexible, biodegradable and long-lasting TENG from silk cocoon waste (Credit: University of Surrey)